

Lecture 9

EE-215 Electronic Devices and Circuits

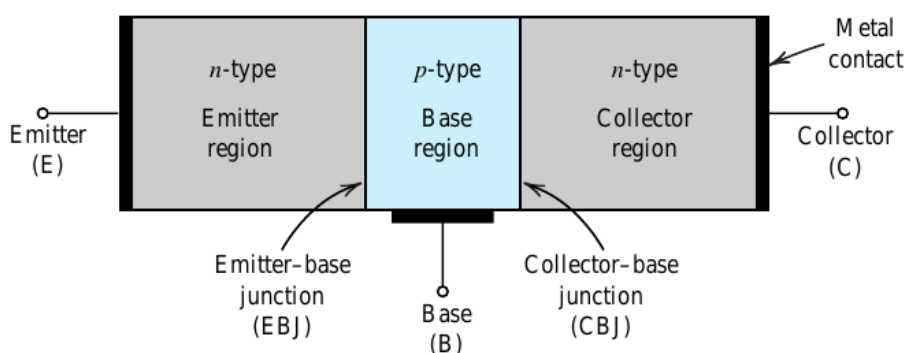
Asst Prof Muhammad Anis Ch

Bipolar Junction Transistors (BJTs)

Device Structure and Physical Operation

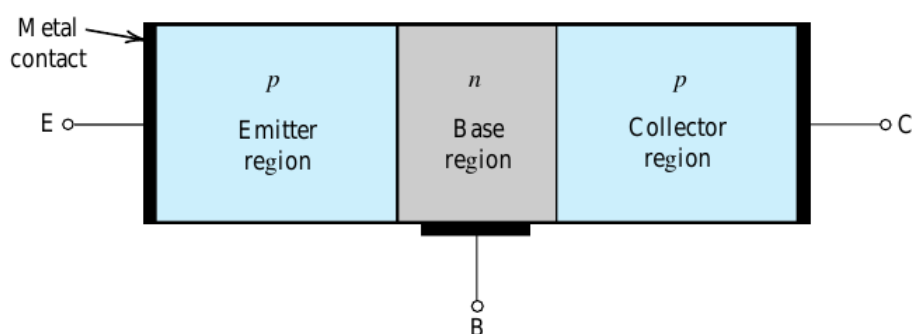
Simplified Structure and Modes of Operation

- fig shows simplified structure of the two types of bipolar junction transistors
 - i.e. npn
 - and pnp



- **Figure 6.1** A simplified structure of the *nnp* transistor.

- the npn bipolar transistor
- contains a thin p-region between two n-regions



- **Figure 6.2** A simplified structure of the *pnp* transistor.

- the pnp bipolar transistor
- contains a thin n-region sandwiched between two p-regions

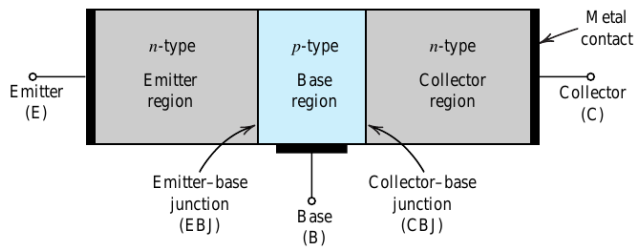


Figure 6.1 A simplified structure of the *nnp* transistor.

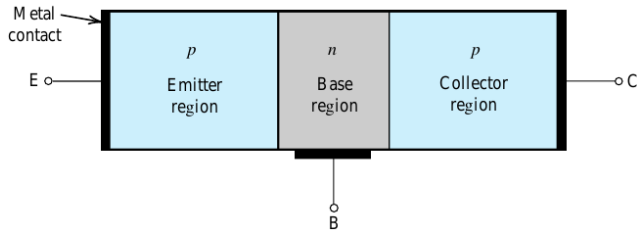


Figure 6.2 A simplified structure of the *pnp* transistor.

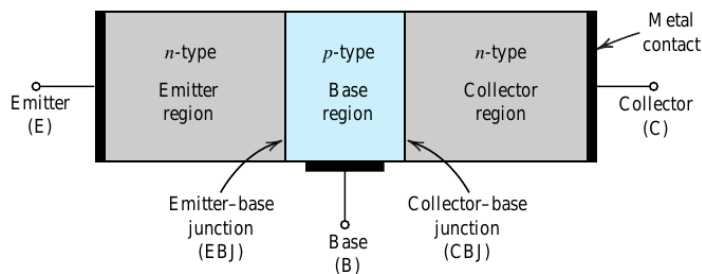
- The three regions and their terminal connections are called the
 - emitter
 - base
 - and collector
- As we will see later, the emitter “emits” charge carriers and
 - the collector “collects” them while the
 - base controls the number of carriers that make this journey
- we readily note from fig, that the device contains two pn junctions,
 - the emitter-base junction (EBJ)
 - and the collector-base junction (CBJ)
- depending upon the bias conditions of each of these junctions,
 - different modes of operation of the BJT are obtained,

Table 6.1 BJT Modes of Operation		
Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward

- also there is a reverse-active mode which occurs when
 - CBJ is forward-biased and EBJ is reverse-biased
 - (but is not of much use)
- The active mode is the one used
 - if the transistor is to operate as an amplifier
- Switching applications (e.g. logic circuits) utilize both
 - the cutoff mode and the saturation mode.
- In the cutoff mode, no current flows because both junctions are reverse-biased.

MOSFET	BJT
Saturation	Active
Triode	Saturation
Cutoff	Cutoff

-
- The active mode is the one used
 - if the transistor is to operate as an amplifier
- Switching applications (e.g. logic circuits) utilize both
 - the cutoff mode and the saturation mode.
- In a bipolar junction transistor (BJT), charges of both polarities
 - i.e. the electrons and holes participate in the current-conduction process,
 - hence the name bipolar
- In contrast, recall that in a MOSFET, current is conducted
 - by charge carriers of only one type,
 - electrons in n-channel devices or
 - holes in a p-channel device
 - Thus MOSFET is a unipolar device
- Note that for a BJT, the two pn junctions are sufficiently close together and are thus called interacting pn junctions
 - the operation of the transistor is therefore totally different from that of two back-to-back diodes



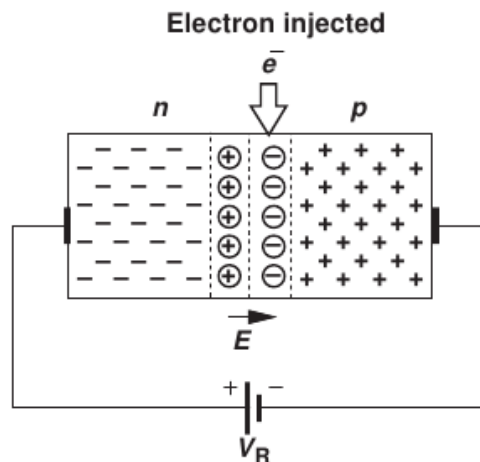
- **Figure 6.1** A simplified structure of the *npn* transistor.
- the operation of a BJT depends on the two pn junctions being in close proximity,
 - so the width of the base must be very narrow, normally in the range of 0.1 to $100\mu\text{m}$

Operation of the npn Transistor in the Active Mode

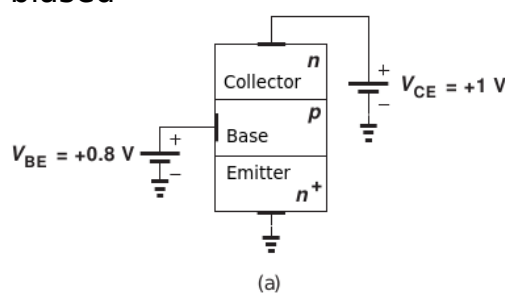
- Among the different modes of operation of the BJT, the active mode is the most important.
 - in this case EBJ is forward-biased, while the CBJ is reverse-biased
 - before we study the operation of BJT in active mode,

- it is important to study an interesting effect in pn junctions

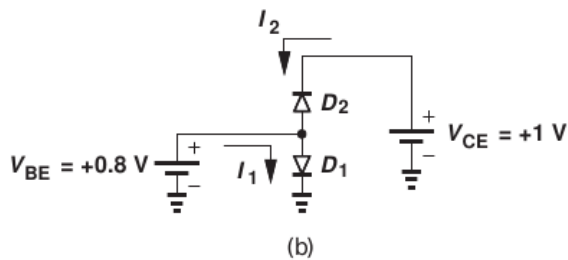
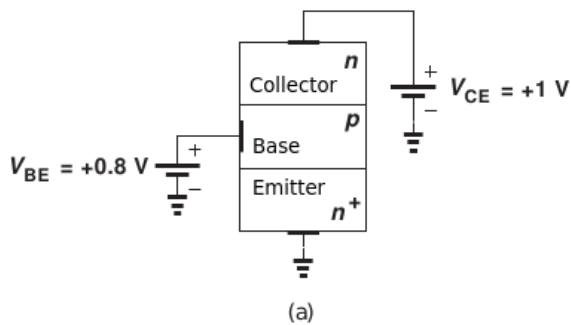
- Consider the reverse-bias pn junction in fig



- recall that the depletion region, sustains a strong electric field
- Now, suppose an electron is somehow injected from outside into the right side of the depletion region
- what will happen to this electron?
- serving as a minority carrier on the p-side, the electron
 - experiences the electric field and is rapidly
 - swept away into the n-side.
 - this ability of a reverse-biased pn junction to efficiently
 - “collect” externally-injected electrons proves essential to the operation of the bipolar transistor
- Let's now consider the operation of the transistor in the active mode
 - in this case EBJ is forward-biased, while the CBJ is reverse-biased

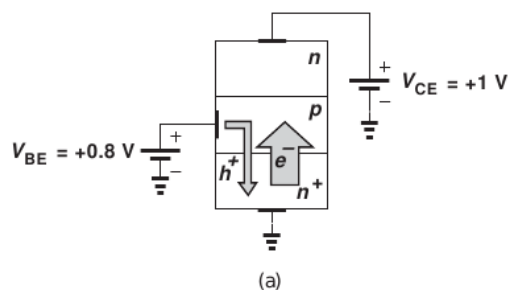


-
- the npn transistor can not be represented as two back-to-back diodes as indicated in fig (b)
 - if the BJT could have been modeled by fig (b), BJT would not act as a transistor ($\because D_1$ conducts and D_2 doesnot)
 - i.e. the current would flow from the base to the emitter but no current through the collector terminal
 - (As the basic transistor principle is that the voltage between two terminals controls the current through the third terminal)

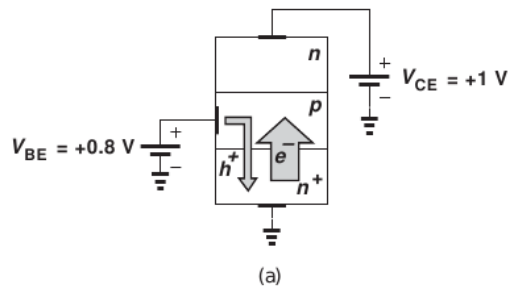


- To see why the BJT cannot be modeled as ,merely two back-to-back diodes,
 - we must examine the flow of charge inside the device.
 - Note that for the BJT, base is very thin.
 - As the EBJ is forward-biased, electrons

- flow from the emitter to the base and
 - holes from the base to the emitter



- for proper transistor operation, the former current
 - must be much greater than the latter
- this is ensured by having emitter doping level
 - to be much higher than that of the base
 - thus the emitter region is denoted as n^+ ,
 - where the superscript exphasizes the high doping level

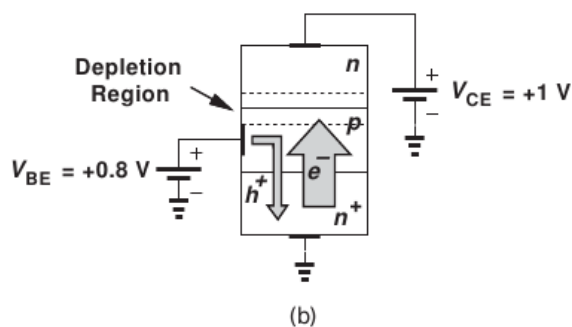
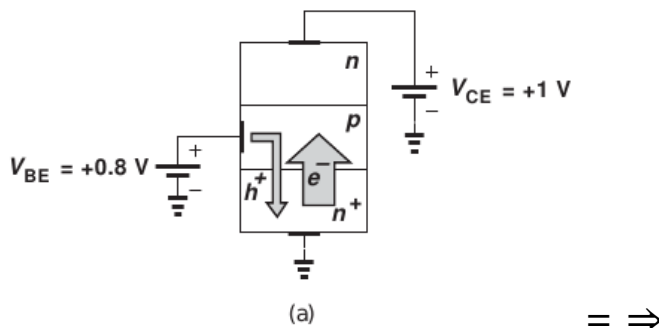


◦ thus, the fig indicates that the

- emitter injects a large number of electrons into the base
- while receiving a small number of holes from the base

• Now the electrons have been injected into the base region.

◦ As the base region is thin, most of the electrons will reach the edge of the collector base depletion region.

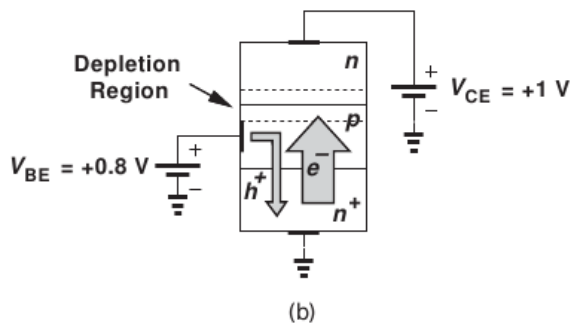


◦ these electrons at the edge of the collector-base depletion

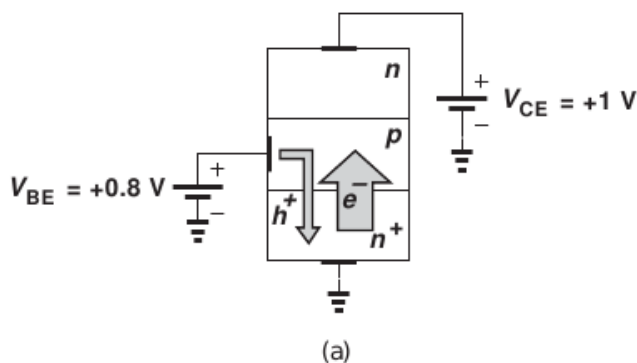
- region, will experience the built-in-electric field and are swept into the collector region

• these electrons at the edge of the collector-base depletion region,

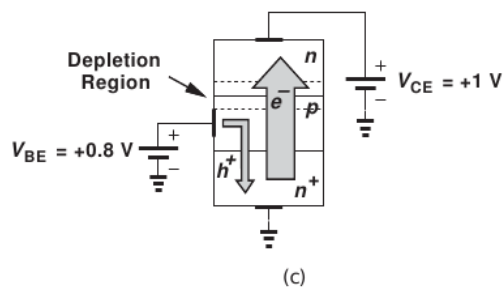
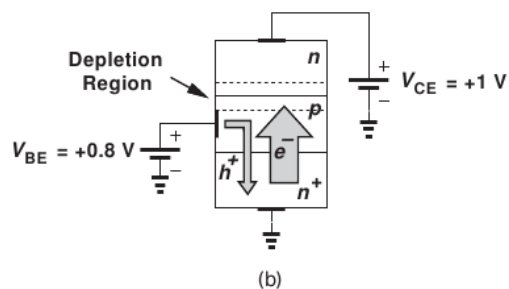
◦ will experience the built-in-electric field and are swept into the collector region



- thus the reverse-biased collector-base junction carries a current
- because minority carriers are “injected” into its depletion region

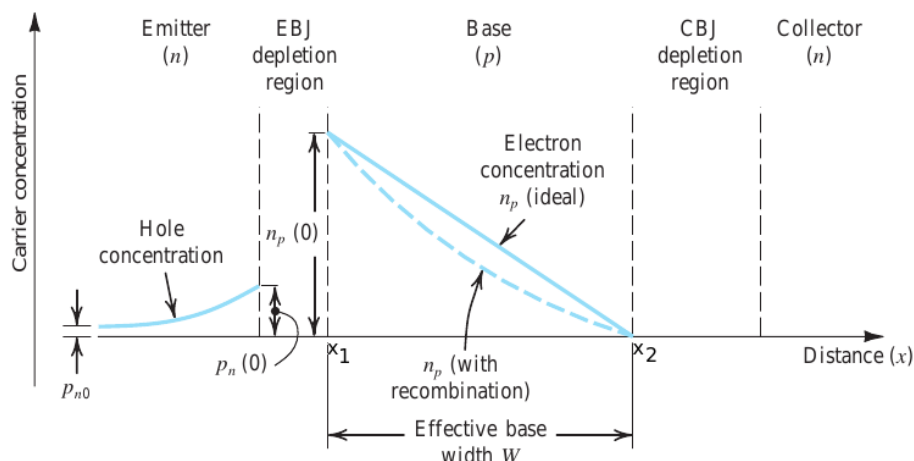


- thus, in the active mode, an npn bipolar transistor
 - carries a large number of electrons from the emitter,
 - though the base, to the collector, while drawing a
 - small amount of holes through the base terminal.



- In the base region, diffusion is the principal mechanism for the flow of electrons injected by the emitter

- the drift current in the base is negligible
- (As the base region acting as a moderate conductor, sustain a small electric field i.e. it allows most of the field to drop across the base-emitter depletion layer)



- from the figure, the density of electrons at $x = x_1$ is very high
 - As any electron arriving at $x = x_2$ is swept away, the density of electrons falls to zero at this point
 - thus, the electron density in the base assumes the profile (a straight line) as shown in fig,
- thus, the electron density in the base assumes the profile (a straight line) as shown in fig,
 - which provides a gradient for the diffusion of electrons
- As for a forward-biased pn junction, the concentration $n_p(0)$ is
 - $n_p(0) \propto e^{v_{BE}/V_T} \Rightarrow n_p(0) = n_{p0} e^{v_{BE}/V_T}$
 - where $n_{p0} = \frac{n_i^2}{N_A}$ (is the proportionality constant)
 - $\Rightarrow n_p(0) = \frac{n_i^2}{N_A} e^{v_{BE}/V_T}$
 - where
 - n_i = the intrinsic carrier density (i.e. the number of free electrons and holes in a unit volume (cm^3) of intrinsic (not doped) silicon at a given temperature
 - N_A = the doping concentration in the base
 - v_{BE} is the forward base-emitter bias voltage
 - V_T is the thermal voltage and $V_T = 25mV$ at room temperature
- because of this minority carrier concentration profile, the electrons
 - injected into the base, diffuse through the base region towards the collector
- using the basic definition of diffusion current density

◦ this electron diffusion current density can be given as

$$\blacksquare J_n = qD_n \frac{dn_p(x)}{dx}$$

■ where q is the magnitude of the electron charge

■ D_n is the diffusion constant (or diffusivity) of electrons in the base

◦ if A_E is the cross-sectional area of the base-emitter junction, then

■ the electron diffusion current I_n can be given as

$$\blacksquare I_n = J_n A_E$$

$$\circ I_n = J_n A_E = A_E q D_n \frac{dn_p(x)}{dx} \therefore J_n = q D_n \frac{dn_p(x)}{dx}$$

$$\bullet I_n = A_E q D_n \frac{dn_p(x)}{dx}$$

$$\circ \text{ from the fig, } \frac{dn_p(x)}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - n_p(0)}{W} = \frac{-n_p(0)}{W}$$

$$\circ = \Rightarrow I_n = A_E q D_n \left(\frac{-n_p(0)}{W} \right)$$

$$\bullet I_n = A_E q D_n \left(\frac{-n_p(0)}{W} \right)$$

$$\circ I_n = A_E q D_n \left(\frac{-n_i^2 e^{v_{BE}/V_T}}{N_A W} \right) \therefore n_p(0) = \frac{n_i^2}{N_A} e^{v_{BE}/V_T}$$

$$\blacksquare I_n = - \left(\frac{A_E q D_n n_i^2}{N_A W} \right) e^{v_{BE}/V_T}$$

$$\blacksquare \text{ or } I_n = -I_S e^{v_{BE}/V_T} \text{ where } I_S = \frac{A_E q D_n n_i^2}{N_A W}$$

◦ I_n is -ve $\Rightarrow$ current flowing from E to C is negative as electrons are flowing from E to C

◦ I_S is called the saturation current

◦ Note that the collector current is the same as I_n and if we assume

■ that the collector current is flowing from the collector to the emitter

$$\blacksquare \text{ then } i_C = -I_n = I_S e^{v_{BE}/V_T}$$

• Some of the electrons that are diffusing through the base region

◦ will combine with holes, which are the majority carriers in the base.

◦ As the base is very thin and lightly doped, the proportion of electrons

■ lost through this recombination process will be small.

◦ this recombination in the base region causes the excess minority

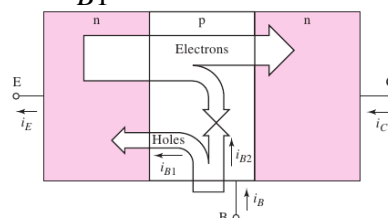
- carrier concentration profile to deviate from a straight line
- and take a slightly concave shape.

The Collector Current

- thus most of the diffusing electrons will reach the boundary of the
 - collector-base depletion region and will be swept across
 - the CBJ depletion region into the collector
 - thus these electrons will be collected to constitute the collector current
 - i.e. $i_C = -I_n = I_S e^{v_{BE}/V_T}$
 - where $I_S = \frac{A_E q D_n n_i^2}{N_A W}$
- Note that, I_S is a strong function of temperature, doubling for every $5^\circ C$ rise in temperature

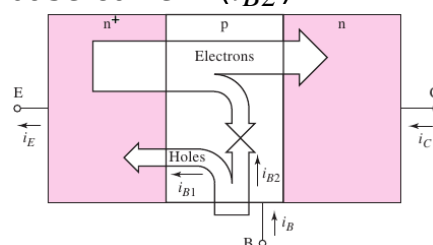
The Base Current

- As the B-E junction is forward-biased, holes from the base are
 - injected across the B-E junction into the emitter.
 - this flow of holes forms one component of the base current
 - This component (i_{B1}) is proportional to e^{v_{BE}/V_T} because of the forward-biased EBJ
 - i.e. $i_{B1} \propto e^{v_{BE}/V_T}$



Electron and hole currents in an npn bipolar transistor biased in the forward-active mode. Emitter, base, and collector currents are proportional to e^{v_{BE}/V_T}

- A few of the electrons (injected from the emitter into the base)
 - recombine with the majority carriers (holes) in the base
 - the holes that are lost must be replaced through the base terminal
 - the flow of such holes is a second component of the base current (i_{B2})



- this “recombination current” is directly proportional to the number of
 - electrons being injected from the emitter (which is an exponential function
 - of the B-E voltage)
 - $\Rightarrow i_{B2} \propto e^{v_{BE}/V_T}$
- thus $i_{B1} \propto e^{v_{BE}/V_T}$ and $i_{B2} \propto e^{v_{BE}/V_T}$
 - As the total base current is the sum of the two components
 - i.e. $i_B = i_{B1} + i_{B2}$
 - $\Rightarrow i_B \propto e^{v_{BE}/V_T}$
 - or $i_B = \left(\frac{1}{\beta}\right) i_C$
 - where $\frac{1}{\beta}$ is the proportionality constant
 - thus $i_B = \frac{i_C}{\beta} = \frac{I_S e^{v_{BE}/V_T}}{\beta}$
 - $\therefore i_C = I_S e^{v_{BE}/V_T}$
- If the concentration of electrons in the n-type emitter is much
 - larger than the concentration of holes in the p-type base,
 - then the number of electrons injected into the base
 - will be much larger than the number of holes
 - injected into the emitter
 - $\Rightarrow i_{B1} < i_C$
- if the base width is small, then the number of electrons that
 - recombine in the base will be small and the i_{B2} component
 - of the base current will also be much smaller than i_C
 - i.e. $i_{B2} < i_C$
 - $\Rightarrow i_B = i_{B1} + i_{B2} < i_C$
 - and as $\beta = \frac{i_C}{i_B} \therefore i_B = \frac{i_C}{\beta}$
 - $\Rightarrow \beta$ will be large
- β is a transistor parameter
 - for modern npn transistors, β is typically in the range of 50 to 200
 - β is called common-emitter current gain
 - thus to obtain high β , both i_{B1} and i_{B2} have to be reduced
 - to reduce i_{B1}
 - emitter is heavily doped, base is lightly doped
 - to reduce i_{B2}
 - base should be thin (i.e. W small)

The Emitter Current

- As the current that enters the transistor must leave it

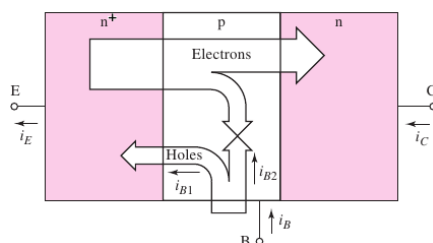
- $\Rightarrow$ the emitter current must be equal to the sum of
 - the collector current i_C and the base current i_B
 - i.e. $i_E = i_C + i_B$
 - as $i_B = \frac{i_C}{\beta}$

- $i_E = i_C + \frac{i_C}{\beta}$

- $i_E = \frac{\beta i_C + i_C}{\beta} = \frac{(\beta+1)i_C}{\beta}$

- $i_E = \frac{1}{\frac{\beta}{\beta+1}} i_C = \frac{1}{\frac{\beta}{\beta+1}} I_S e^{v_{BE}/V_T}$

- or $i_E = \frac{i_C}{\alpha} = \frac{I_S e^{v_{BE}/V_T}}{\alpha}$



- where $\alpha = \frac{\beta}{\beta+1}$, (α is called the common-base current gain)
- $i_E = i_C + i_B$, $i_E = \frac{i_C}{\alpha} = \frac{I_S e^{v_{BE}/V_T}}{\alpha}$
 - where $\alpha = \frac{\beta}{\beta+1}$
 - α is called the common-base current gain
 - α is less than but very close to unity
 - e.g. if $\beta = 100 \Rightarrow \alpha = \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.99$
- $\alpha = \frac{\beta}{\beta+1} \Rightarrow \alpha(\beta+1) = \beta$ or $\alpha\beta + \alpha = \beta$
 - $\alpha\beta - \beta = -\alpha \Rightarrow \beta(\alpha - 1) = -\alpha$
 - or $\beta = \frac{-\alpha}{\alpha-1} = \frac{\alpha}{1-\alpha}$
- thus $\alpha = \frac{\beta}{\beta+1}$, $\beta = \frac{\alpha}{1-\alpha}$

thus as a summary,

- $i_C = I_S e^{v_{BE}/V_T}$
- $i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$
- $i_E = i_C + i_B$, $i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$
 - where $\alpha = \frac{\beta}{\beta+1}$, $\beta = \frac{\alpha}{1-\alpha}$

Operation of the npn Transistor in the Active Mode

Equivalent Circuit Models

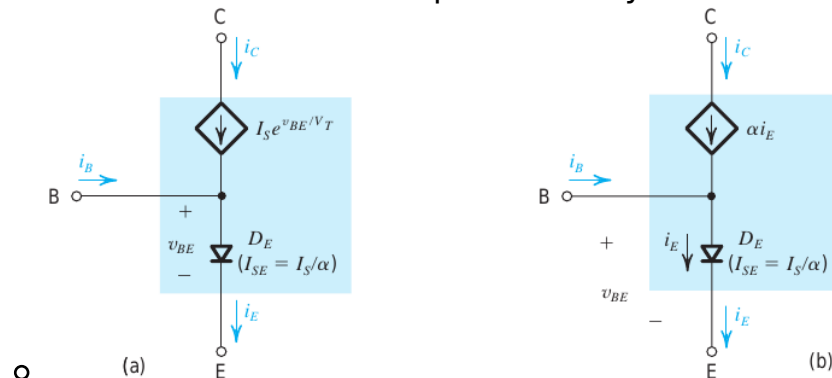
- thus when, an npn BJT is in active mode,
 - the forward bias voltage v_{BE} causes an exponentially
 - related current i_C to flow in the collector terminal
 - i.e. $i_C = I_S e^{v_{BE}/V_T}$
 - this collector current is independent of the value of the collector voltage
 - (ignoring the base-width modulation effect or Early effect),
 - as long as the CBJ remains reverse biased i.e. $v_{CB} \geq 0$
 - thus in the active mode, the collector terminal behaves as
 - an ideal constant-current source where the value
 - of current is determined by v_{BE}
 - $i_C = I_S e^{v_{BE}/V_T}$ i.e. i_C is determined by v_{BE}

- $i_C = I_S e^{v_{BE}/V_T}$

- $i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$

- $i_E = i_C + i_B$, $i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$

- this behaviour can be represented by the models in fig



- $i_C = I_S e^{v_{BE}/V_T}$, $i_E = i_C + i_B$, $i_E = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$

- from fig $i_C = I_S e^{v_{BE}/V_T}$, the diode D_E has a scale current of $I_{SE} = I_S/\alpha$

- $\Rightarrow$ for D_E , $i_{DE} = \frac{I_S}{\alpha} e^{v_{BE}/V_T} = i_E$

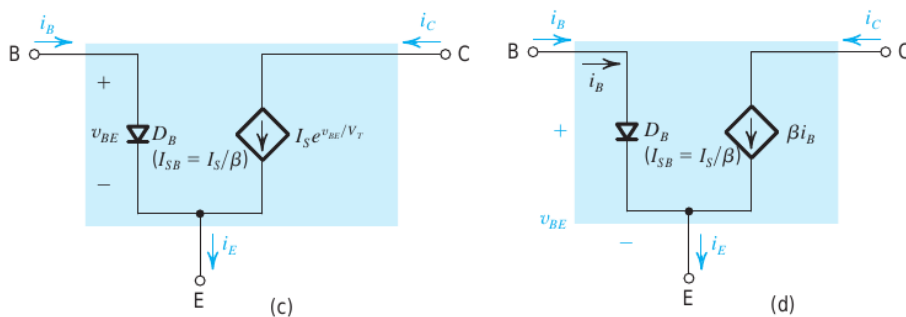
- also from fig, $i_B + i_C = i_E$

- $i_C = I_S e^{v_{BE}/V_T}$

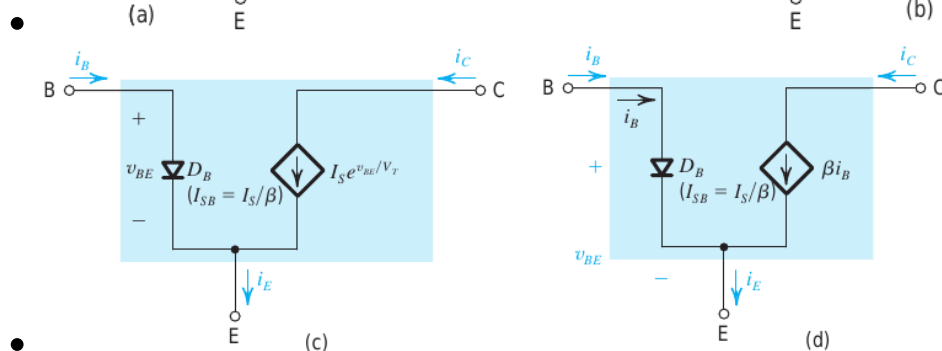
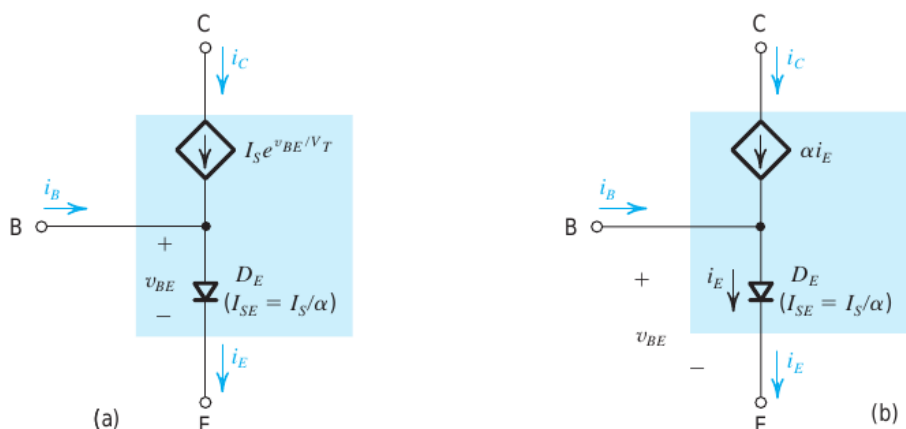
- $i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$

- $i_E = i_C + i_B$, $i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$

◦ this behaviour can also be represented by the models in fig



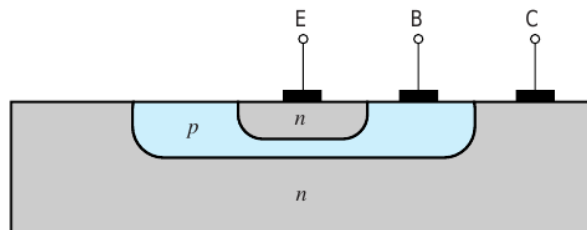
- $i_C = I_S e^{v_{BE}/V_T}$, $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T}$, $i_E = i_C + i_B$
 - from fig $i_C = I_S e^{v_{BE}/V_T}$, the diode D_B has a scale current of $I_{SB} = I_S/\beta$
 - $\Rightarrow$ for D_B , $i_{DB} = \frac{I_S}{\beta} e^{v_{BE}/V_T} = i_B$
 - also from fig, $i_E = i_B + i_C$



- Note that these models are non-linear because of the exponential relationship of the
 - current i_E through diode D_E and the voltage v_{BE} (in case of fig (a) and (b))
 - current i_B through diode D_B and the voltage v_{BE} (in case of fig (c) and (d))
 - also note that these models are valid for any positive value of v_{BE} (provided the BJT is in active mode)
 - and are therefore called large signal models

Structure of Actual Transistors

- A more realistic (still simplified though) cross-section of an npn BJT is shown in fig



- **Figure 6.7** Cross-section of an npn BJT.
 - Note that the device is not symmetrical
 - This asymmetry occurs because the geometries of the emitter and collector
 - regions are not the same, and the impurity doping concentration in the
 - emitter (n^+) , base (p) and collector (n) are different.
 - thus even though the emitter and collector both are of n-type.
 - switching them makes the device behave in drastically different ways
 - if the emitter and collector are reversed i.e. if CBJ is forward-biased
 - and the EBJ is reverse-biased, the device is said to operate in the reverse-active mode
 - the resulting values of α_R and β_R (with R denoting reverse) are much
 - lower than the values of α and β , obtained in the “forward” active mode.
 - thus the reverse active mode is not of much significance
 - the structure in fig also indicates that the CBJ has a much larger area than the EBJ
 - thus the CB diode D_C has a saturation current I_{SC} that is much larger
 - than the saturation current (I_{SE}) of the EB diode D_E
 - typically I_{SC} is 20 to 100 times larger than I_{SE} (recall that $I_{SE} = \frac{I_S}{\alpha}$)
-